

OpenStax Calculus Volume 3

Section 4.6 Exercises: Directional Derivatives and the Gradient

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For the following exercises, find the directional derivative using the limit definition only.

1. $f(x, y) = 5 - 2x^2 - \frac{1}{2}y^2$ at point $P(3, 4)$ in the direction of $\mathbf{u} = (\cos \frac{\pi}{4})\mathbf{i} + (\sin \frac{\pi}{4})\mathbf{j}$
2. $f(x, y) = y^2 \cos(2x)$ at point $P(\frac{\pi}{3}, 2)$ in the direction of $\mathbf{u} = (\cos \frac{\pi}{4})\mathbf{i} + (\sin \frac{\pi}{4})\mathbf{j}$
3. Find the directional derivative of $f(x, y) = y^2 \sin(2x)$ at point $P(\frac{\pi}{4}, 2)$ in the direction of $\mathbf{u} = 5\mathbf{i} + 12\mathbf{j}$.

For the following exercises, find the directional derivative of the function at point P in the direction of $\mathbf{u}$ or $\mathbf{v}$ as appropriate.

4. $f(x, y) = xy$, $P(0, -2)$, $\mathbf{v} = \frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j}$
5. $h(x, y) = e^x \sin y$, $P(1, \frac{\pi}{2})$, $\mathbf{v} = -\mathbf{i}$
6. $h(x, y, z) = xyz$, $P(2, 1, 1)$, $\mathbf{v} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$
7. $f(x, y) = xy$, $P(1, 1)$, $\mathbf{u} = \langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \rangle$
8. $f(x, y) = x^2 - y^2$, $\mathbf{u} = \langle \frac{\sqrt{3}}{2}, \frac{1}{2} \rangle$, $P(1, 0)$
9. $f(x, y) = 3x + 4y + 7$, $\mathbf{u} = \langle \frac{3}{5}, \frac{4}{5} \rangle$, $P(0, \frac{\pi}{2})$
10. $f(x, y) = e^x \cos y$, $\mathbf{u} = \langle 0, 1 \rangle$, $P = (0, \frac{\pi}{2})$
11. $f(x, y) = y^{10}$, $\mathbf{u} = \langle 0, -1 \rangle$, $P = (1, -1)$
12. $f(x, y) = \ln(x^2 + y^2)$, $\mathbf{u} = \langle \frac{3}{5}, \frac{4}{5} \rangle$, $P(1, 2)$
13. $f(x, y) = x^2y$, $P(-5, 5)$, $\mathbf{v} = 3\mathbf{i} - 4\mathbf{j}$
14. $f(x, y, z) = y^2 + xz$, $P(1, 2, 2)$, $\mathbf{v} = \langle 2, -1, 2 \rangle$

For the following exercises, find the directional derivative of the function in the direction of the unit vector $\mathbf{u} = \cos \theta \mathbf{i} + \sin \theta \mathbf{j}$.

15. $f(x, y) = x^2 + 2y^2$, $\theta = \frac{\pi}{6}$
16. $f(x, y) = \frac{y}{x+2y}$, $\theta = -\frac{\pi}{4}$

17. $f(x, y) = \cos(3x + y), \theta = \frac{\pi}{4}$
18. $w(x, y) = ye^x, \theta = \frac{\pi}{3}$
19. $f(x, y) = x \arctan(y), \theta = \frac{\pi}{2}$
20. $f(x, y) = \ln(x + 2y), \theta = \frac{\pi}{3}$

For the following exercises, find the gradient.

21. Find the gradient of $f(x, y) = \frac{14-x^2-y^2}{3}$. Then, find the gradient at point $P(1, 2)$.
22. Find the gradient of $f(x, y, z) = xy + yz + xz$ at point $P(1, 2, 3)$.
23. Find the gradient of $f(x, y, z)$ at P and the directional derivative in the direction of $\mathbf{u}$:
 $f(x, y, z) = \ln(x^2 + 2y^2 + 3z^2), P(2, 1, 4), \mathbf{u} = \frac{-3}{13}\mathbf{i} - \frac{4}{13}\mathbf{j} - \frac{12}{13}\mathbf{k}$.
24. $f(x, y, z) = 4x^5y^2z^3, P(2, -1, 1), \mathbf{u} = \frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$

For the following exercises, find the directional derivative of the function at point P in the direction of Q .

25. $f(x, y) = x^2 + 3y^2, P(1, 1), Q(4, 5)$
26. $f(x, y, z) = \frac{y}{x+z}, P(2, 1, -1), Q(-1, 2, 0)$

For the following exercises, find the derivative of the function at P in the direction of $\mathbf{u}$.

27. $f(x, y) = -7x + 2y, P(2, -4), \mathbf{u} = 4\mathbf{i} - 3\mathbf{j}$
28. $f(x, y) = \ln(5x + 4y), P(3, 9), \mathbf{u} = 6\mathbf{i} + 8\mathbf{j}$
29. [T] Use technology to sketch the level curve of $f(x, y) = 4x - 2y + 3$ that passes through $P(1, 2)$ and draw the gradient vector at P .
30. [T] Use technology to sketch the level curve of $f(x, y) = x^2 + 4y^2$ that passes through $P(-2, 0)$ and draw the gradient vector at P .

For the following exercises, find the gradient vector at the indicated point.

31. $f(x, y) = xy^2 - yx^2, P(-1, 1)$
32. $f(x, y) = xe^y - \ln(x), P(3, 0)$
33. $f(x, y, z) = xy - \ln(z), P(2, -2, 2)$
34. $f(x, y, z) = x\sqrt{y^2 + z^2}, P(-2, -1, -1)$

For the following exercises, find the derivative of the function.

35. $f(x, y) = x^2 + xy + y^2$ at point $(-5, -4)$ in the direction the function increases most rapidly
36. $f(x, y) = e^{xy}$ at point $(6, 7)$ in the direction the function increases most rapidly
37. $f(x, y) = \arctan(\frac{y}{x})$ at point $(-9, 9)$ in the direction the function increases most rapidly
38. $f(x, y, z) = \ln(xy + yz + zx)$ at point $(-9, -18, -27)$ in the direction the function increases most rapidly

39. $f(x, y, z) = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$ at point $(5, -5, 5)$ in the direction the function increases most rapidly

For the following exercises, find the maximum rate of change of f at the given point and the direction in which it occurs.

40. $f(x, y) = xe^{-y}$, $(1, 0)$
 41. $f(x, y) = \sqrt{x^2 + 2y}$, $(4, 10)$
 42. $f(x, y) = \cos(3x + 2y)$, $(\frac{\pi}{6}, -\frac{\pi}{8})$

For the following exercises, find equations of

- (a) the tangent plane and
 (b) the normal line to the given surface at the given point.

43. The level surface $f(x, y, z) = 12$ for $f(x, y, z) = 4x^2 - 2y^2 + z^2$ at point $(2, 2, 2)$.
 44. $f(x, y, z) = xy + yz + xz = 3$ at point $(1, 1, 1)$
 45. $f(x, y, z) = xyz = 6$ at point $(1, 2, 3)$
 46. $f(x, y, z) = xe^y \cos z - z = 1$ at point $(1, 0, 0)$

For the following exercises, solve the problem.

47. The temperature T in a metal sphere is inversely proportional to the distance from the center of the sphere (the origin: $(0, 0, 0)$). The temperature at point $(1, 2, 2)$ is 120°C .
 (a) Find the rate of change of the temperature at point $(1, 2, 2)$ in the direction toward point $(2, 1, 3)$.
 (b) Show that, at any point in the sphere, the direction of greatest increase in temperature is given by a vector that points toward the origin.
48. The electrical potential (voltage) in a certain region of space is given by the function $V(x, y, z) = 5x^2 - 3xy + xyz$.
 (a) Find the rate of change of the voltage at point $(3, 4, 5)$ in the direction of the vector $\langle 1, 1, -1 \rangle$.
 (b) In which direction does the voltage change most rapidly at point $(3, 4, 5)$?
 (c) What is the maximum rate of change of the voltage at point $(3, 4, 5)$?
49. If the electric potential at a point (x, y) in the xy -plane is $V(x, y) = e^{-2x} \cos(2y)$, then the electric intensity vector at (x, y) is $\mathbf{E} = -\nabla V(x, y)$.
 (a) Find the electric intensity vector at $(\frac{\pi}{4}, 0)$.
 (b) Show that, at each point in the plane, the electric potential decreases most rapidly in the direction of the vector $\mathbf{E}$.
50. In two dimensions, the motion of an ideal fluid is governed by a velocity potential ϕ . The velocity components of the fluid u in the x -direction and v in the y -direction, are given by $\langle u, v \rangle = \nabla \phi$. Find the velocity components associated with the velocity potential $\phi(x, y) = \sin(\pi x) \sin(2\pi y)$.